Vijay Sundar Nagamalla

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AI Application Developer | Full Stack Engineer | Agentic Systems Architect

PROFESSIONAL SUMMARY

Veteran L&D professional with 8+ years of experience delivering and operating high-scale technical training programs. I have mentored 50,000+ learners in core programming (C, Java, Python, SQL, and DSA) across 40+ institutions while managing complex learning operations and LMS ecosystems. Driven by a passion for self-growth and technological innovation, I also architect production-grade AI solutions, including RAG pipelines and agentic frameworks, to solve real-world automation challenges.

TECHNICAL SKILLS

AI / ML: RAG Pipelines, MCP (Model Context Protocol), Claude API, OpenAI API, NVIDIA NIM, Whisper ONNX, DeepSeek

Frontend: React 19, TanStack Start, TypeScript, TailwindCSS, Framer Motion, TanStack Query, shadcn/ui

Backend: FastAPI (Python), Node.js, C# / .NET 8 WPF, REST APIs, JWT, RBAC, Judge0

Databases: PostgreSQL, Redis, Supabase, SQLite

DevOps: Docker, Nginx, Vercel, GitHub Pages, GitHub Actions

Languages: Python, JavaScript, TypeScript, C#, Java, SQL

Training and Ops: Technical Training, Power BI, Google Sheets API, MS Excel, LMS, Learning KPIs

PROFESSIONAL EXPERIENCE

Senior Technical Trainer & Learning Operations / AI Developer May 2020 – Present

Freelance

- Architected and deployed 7+ end-to-end AI and full-stack platforms, including EdTech analytics (Almaanac, Vukti), Salesforce tracking, and an AI-powered stealth desktop assistant (Forward)
- Delivered high-impact technical training in Java, Python, SQL, and DSA to 25,000+ learners, achieving ~50% average assessment improvement
- Managed end-to-end learning operations: from multi-track session scheduling and batch planning to learner enrollment and attendance tracking
- Partnered with instructional designers and stakeholders to ensure seamless program execution and on-time delivery of learning modules
- Leveraged MS Excel and Power BI to track real-time KPIs, monitor learner completion rates, and maintain operational data with 95%+ accuracy

Technical Project Manager May 2023 – December 2024

Xplore.co.in

- Spearheaded large-scale upskilling programs as Lead Trainer and Project Owner, ensuring 100% on-time delivery of technical training and labs
- Built and maintained interactive Power BI dashboards for leadership to monitor KPIs, learner engagement, and program health in real-time
- Orchestrated end-to-end program operations, including batch planning, stakeholder coordination, and multi-track logistics
- Streamlined training workflows and collaborated with instructional designers to optimize content delivery and assessment results

Associate Trainer June 2018 – May 2020

FACE

- Facilitated high-impact technical training in Aptitude, C, Java, Python, and SQL for batches of 80+ learners
- Developed comprehensive lesson plans, presentations, and hands-on assessments to drive technical proficiency and student outcomes
- Managed end-to-end classroom operations, from attendance tracking to performance evaluation and progress reporting
- Provided structured feedback and personalized learning support to enhance student technical skills and interview readiness

KEY PROJECTS

Almaanac — Classroom Coding Analytics Platform Click here to open

TanStack Start (React 19), TanStack Router, TanStack Query v5, Supabase, Tailwind CSS v4, Recharts, TypeScript 5.8, Vercel

- Built a full-stack classroom analytics platform that monitors LeetCode activity across classrooms and cohorts for educators and placement officers
- LeetCode GraphQL integration: scrapes solved counts, submission calendars, contest ratings, badges, and language stats with exponential backoff

- Dashboards with leaderboards, behavioral buckets ("Active Today", "At Risk", "Top Performers"), heatmaps, 30-day trends, and difficulty-split charts
- Automated refresh: daily Vercel cron + GitHub Actions pump with Postgres-level lease locking and rate-limited concurrency

Vukti — Unified Academic Intelligence Platform
React 19, Vite 7, TailwindCSS 4, FastAPI, PostgreSQL 16, Redis 7, Docker, Judge0, NVIDIA AI

- Architecting a 9-module student operating system for technical education (B2B + B2C), targeting 10,000 concurrent users
- Multi-platform sync from 6 coding platforms (LeetCode, HackerRank, CodeChef, Codeforces, GFG, HackerEarth) with async scraper orchestrator
- Assessment engine: MCQ + Coding + SQL with Docker-isolated Judge0 execution, full-screen proctoring, and tab-blur detection
- VIMY gamified practice: 1,000+ problems, streaks, coins, XP levels, badges, leagues, and activity heatmaps

Salesforce Tracker — Trailhead Profile Scraper Click here to open
TanStack Start (React 19), TanStack Router, Supabase, Tailwind CSS v4, shadcn/ui, XLSX, TypeScript

- Bulk-scrape Salesforce Trailhead profiles from URLs or Excel uploads for institutional tracking and certification monitoring
- Auto-detect Trailhead profile columns, batch-scrape with progress tracking, and merge with uploaded metadata
- Persistent session history with export to Excel and server-side scraping via TanStack Start API routes

FDE Flow Atlas — Interactive Learning Platform Click here to open
Vanilla HTML/CSS/JS, SVG Animations, LocalStorage Progress, GitHub Pages

- 133-topic interactive curriculum for Forward Deployed Engineering with animated SVG flow diagrams, case studies, and 11 hands-on projects
- Step-by-step narration with code highlighting, playback controls, interactive quizzes, and fill-in templates
- Covers RAG pipelines, tool calling, agentic loops, case decomposition — all progress tracked via localStorage

ILFU — AI Research Project GitHub
Python, AI/ML Stack

- Experimental AI research and experimentation platform currently under active development

Forward — AI Class Assistant (Desktop)
C# WPF, .NET 8, Claude API, DeepSeek, NVIDIA NIM, NAudio WASAPI, Whisper ONNX, AMD XDNA NPU

- Built an invisible-to-screen-capture AI desktop assistant with stealth overlay (WDA_EXCLUDEFROMCAPTURE P/Invoke)
- Multi-provider AI fallback chain: Claude → DeepSeek → NVIDIA NIM → Groq → Ollama with automatic failover
- Hardware-optimized: Whisper STT on AMD XDNA NPU, embeddings on Radeon DirectML GPU, Silero VAD on CPU
- Hotkey-driven UX: Alt+Space (toggle overlay), Ctrl+Space (explain last 30s), Alt+A (focus input)

CORE COMPETENCIES

AI & Desktop Engineering: Claude/DeepSeek/NVIDIA NIM integration, Whisper STT, on-device AI, stealth desktop UX

Full Stack Engineering: React 19, TanStack Start, FastAPI, PostgreSQL, Docker, C# WPF / .NET 8

EdTech Platforms: Classroom analytics, assessment engines, multi-platform coding trackers, gamified learning

Technical Training: Java, Python, SQL, DSA, 50,000+ learners trained across 40+ colleges

Learning Operations: LMS, KPI Reporting, Stakeholder Management, Curriculum Design